



Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

An Explainable Machine Learning Framework for Binary Classification Using Real-World Data

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Objective

Understand

Examine how SHAP and LIME explain ML model predictions in binary classification across healthcare, finance, and fraud detection domains.

Study

Investigate the relationship between predictive accuracy and explanation quality across three real-world benchmark datasets.

Compare

Analyze differences in explanation stability and faithfulness between inherently interpretable models and black-box classifiers.

Explore

Understand how explanation consistency and stability vary across models, datasets, and random seeds.

Introduction

- Machine learning has evolved from linear classifiers to deep architectures capable of superhuman performance across a wide range of tasks.
- Binary classification is practically critical: medical diagnostics (disease/no-disease), finance (default/no-default), and fraud detection.
- Black-box models trade interpretability for performance, raising urgent questions about transparency in high-stakes decisions.
- Regulatory frameworks (GDPR, ECOA) increasingly mandate explainability, making XAI a practical and legal necessity.

Introduction

Inherently Interpretable

- Merit-based transparency
- Logistic Regression coefficients directly quantify feature influence
- Decision Trees provide human-readable paths
- Limited predictive power on complex data

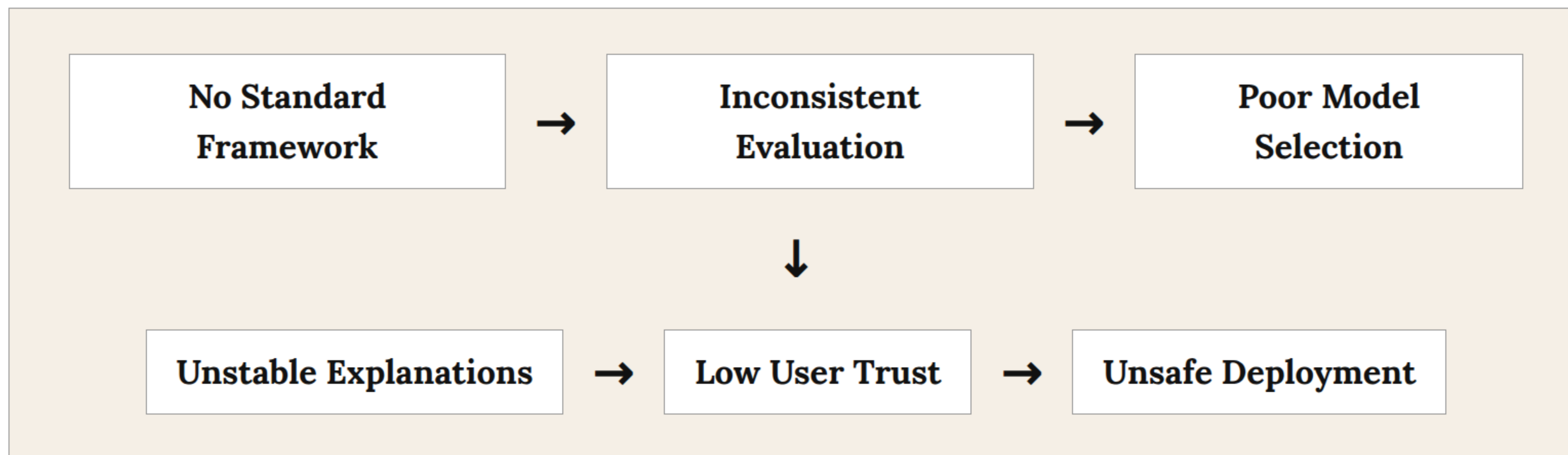
Post-Hoc XAI Methods

- Applied to already-trained black-box models
- SHAP: Theoretically grounded via Shapley values; global and local explanations
- LIME: Model-agnostic locally faithful linear surrogates
- Combines full model power with interpretability

Problem Statement

- No common, reproducible framework exists to assess both prediction accuracy and explanation quality in real-world binary classification settings.
- Most prior work uses only accuracy as evaluation metric, or proposes a new explanation method without rigorous comparison.
- Explanation consistency and stability are rarely studied systematically across models and datasets.
- Transparency and principled model selection for high-stakes deployment remain unsolved challenges.

Problem Statement



Methodology

1 Data Acquisition and Preprocessing

Three datasets: UCI Adult Income (48,842), Pima Diabetes (768), German Credit (1,000). Imputation, one-hot encoding, z-score normalization, SMOTE on training set only.

2 Feature Engineering and Selection

Variance thresholding eliminates low-variance features; mutual information scoring selects top-k features via cross-validation.

3 Model Training and Hyperparameter Optimization

Four classifiers: LR, RF, GBM (XGBoost), SVM (RBF). Trained via stratified 5-fold cross-validation with grid search.

4 Performance Evaluation and Explanation Generation

Metrics: Accuracy, AUC-ROC, Precision, Recall, F1. SHAP and LIME applied for global/local explanations; faithfulness and stability assessed.

Python Validation

Python Simulation

Python 3.10 on Intel i7-12700, 32 GB RAM, RTX 3060.
scikit-learn 1.3, XGBoost 2.0, SHAP 0.42, LIME 0.2,
pandas 2.0.

Modular Pipeline

Preprocessing, training, and explanation modules are independently configurable. Results serialized as JSON and images for reproducibility via Git.

Model Implementation

LR, RF, GBM, SVM implemented with parameterized grid search. SHAP via TreeExplainer (RF, GBM) and KernelExplainer (LR, SVM). LIME with 500 perturbed samples per instance.

Ranking Comparison & Validation

Models retrained across 10 random seeds. Kendall's tau above 0.80 indicates high stability. SHAP-LIME Spearman correlation 0.73 across all models and datasets.

Key Findings

- GBM achieved the highest AUC-ROC: 0.927 on Adult Income and 0.891 on German Credit. Random Forest led marginally on Pima Diabetes (0.846 vs 0.841).
- SHAP provided more stable feature attributions than LIME. GBM had the highest Kendall's tau (0.88); SVM the lowest (0.71).
- SHAP consistently identified glucose, BMI, and age as the top three features across all models on Pima Diabetes — aligned with medical literature.
- Average Spearman correlation between SHAP and LIME was 0.73. Agreement dropped significantly near decision boundaries.
- Logistic Regression remained competitive on linearly separable data, achieving AUC-ROC of 0.896 on the Adult Income dataset.

Conclusion

Modular Framework

The XMLF allows seamless integration of new datasets, classifiers, and interpretive methodologies — enabling plug-and-play experimentation without rebuilding the pipeline.

Stability as Primary Criterion

Explanation stability must be placed on the same level as AUC-ROC and accuracy. Accurate models with inconsistent explanations are unsuitable for high-stakes deployment.

Domain-Contingent Quality

Explanation quality depends on domain characteristics. High feature correlation and class imbalance create greater challenges for generating consistent explanations.

Practical Recommendation

GBM (XGBoost) with SHAP explanations is recommended for tabular binary classification requiring both high predictive performance and stable interpretations.

Future Scope

1**Counterfactual Explanations**

Extend XMLF with counterfactual algorithms to show users the minimal changes that would alter a prediction, improving actionability beyond feature attribution.

2**Algorithmic Fairness Evaluation**

Integrate fairness metrics (demographic parity, equalized odds) and study how SMOTE-based resampling affects fairness across protected attributes.

3**Interactive UI for Domain Experts**

Build an interface enabling clinicians and credit analysts to review, validate, and contest model explanations without ML expertise.

4**Extension to Non-Tabular Data**

Extend the framework beyond structured tabular data to image, text, and sequential data modalities.



Thank You